

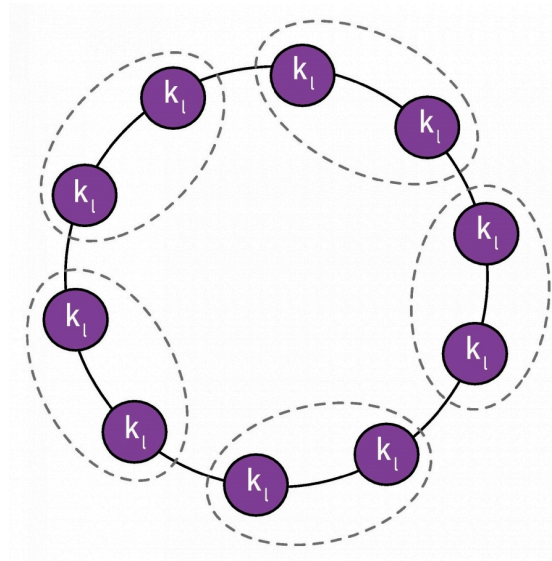
L7: Modularity Resolution

You may be wondering: is the modularity metric always a reliable way to choose a community assignment? Does it ever fail to point us in the right direction?

The answer is yes. The modularity metric cannot be used to discover very small communities, relative to the size of the given network.

To see why recall the formula we derived earlier for the change in modularity after merging two communities A and B:

$$\Delta M_{AB} = \frac{l_{AB}}{L} - \frac{\kappa_A \kappa_B}{2L^2}$$



A partition in which pairs of neighboring cliques are merged into a single community, as indicated by the dotted lines. (Image 9.34) from networksciencebook.com

Suppose that a network consists of multiple communities, as shown in the visualization, with only a single link between any two communities. Clearly, a good community detection algorithm should not merge any of these communities

To simplify, suppose that the total degree of each community is κ_l . Based on the previous formula, if we merge any two of these connected communities the modularity difference will be:

$$\Delta M_{AB} = \frac{1}{L} - \frac{\kappa_l^2}{2L^2} = \frac{1}{L} \left(1 - \frac{\kappa_l^2}{2L}\right)$$

So, the merging of the two small communities will take place if $\Delta M_{AB} > 0$ which is equivalent to $\kappa_l < \sqrt{2L}$

In other words, a modularity maximization algorithm will always merge two communities if the total degree of each of those communities is smaller than the critical value $\sqrt{2L}$.

This critical value is referred to as modularity resolution because it is the smaller community size (*in terms of total degree*) that a modularity maximization algorithm can detect.

For example, for a network with a million links, the modularity resolution is 1414, which means that any two connected communities (*even with a single link*) with total degree less than this limit will be merged, and so they will not be correctly identified.

Food For Thought

How would you try to avoid the modularity resolution issue -- and try to discover even small communities?